

Grade 8
Unit 7
Lesson 2 Practice

Name Answer key
Date _____
Period _____

The community of Siesta Key, Florida has the fastest shrinking population in the state. In 2010 the population of Siesta Key was 6,568. Then in 2019, the population was 5,570. It was determined that in 2021, the linear pattern continued.

1. Calculate the rate of change of the Siesta Key population.

$$\frac{5570-6568}{9} = \frac{-998}{9}$$

2. Determine whether the rate of change represents an increase or decrease in the population of Siesta Key.

decrease

3. Based on the answers to the previous questions, what is the slope of this linear relationship?

$$\frac{-998}{9}$$

4. Complete the statement to interpret the slope as the rate of change for this context.

In this context, the slope means that the population of Siesta Key

☐ increases	by	☐ 5,570	☐ year	per	☒ year.
☒ decreases		☐ 998			☒ people
		☒ 111			

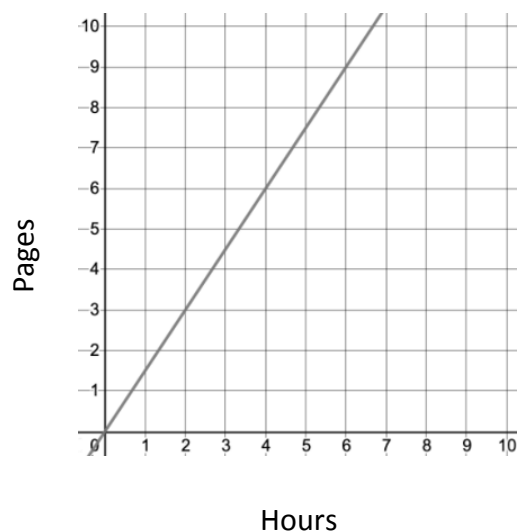
The graph shows how the length of Gabby's book report depends on the number of hours she spends writing.

5. Find the slope of the line.

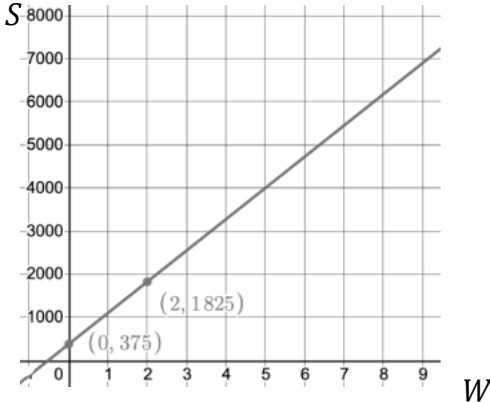
$$\frac{6-0}{4-0} = \frac{6}{4} = \frac{3}{2} = 1.5$$

6. Interpret the slope for this context.

Gabby writes 1.5 pages per hour.



For each linear relationship, determine the rate of change and interpret its meaning in the given context:

Linear Relationship	Rate of Change	Interpretation										
7. Sarah started a savings account, the graph represents the relationship between the amount of money in the savings account, S, and the number of weeks W. 	$\frac{1825-375}{2-0} = \frac{1450}{2} = \$725/\text{Week}$	Sarah saved \$725 per week.										
8. The cost of a large cheese pizza is \$12.50 and each additional topping costs \$0.75.	\$0.75/Topping	The cost of the pizza increases \$0.75 per topping.										
9. We left for our vacation at 11:00AM the table indicates our mileage each hour during our trip. <table border="1" data-bbox="276 1312 613 1509"><thead><tr><th>Time</th><th>Mileage</th></tr></thead><tbody><tr><td>11:00</td><td>13423</td></tr><tr><td>12:00</td><td>13498</td></tr><tr><td>1:00</td><td>13573</td></tr><tr><td>2:00</td><td>13648</td></tr></tbody></table>	Time	Mileage	11:00	13423	12:00	13498	1:00	13573	2:00	13648	$\frac{13498-13423}{12-11} = 75 \text{ miles/hour}$	We traveled 75 miles per hour.
Time	Mileage											
11:00	13423											
12:00	13498											
1:00	13573											
2:00	13648											
10. Nathan is planning a birthday party. The cost for 8 people is \$123 and the cost for 12 people is \$167.	$\frac{167-123}{12-8} = \frac{44}{4} = \$11/\text{Person}$	The cost was \$11 per person.										